Credit Default Risk Prediction

Access: https://github.com/mandypeng-brown/data1030-project-credit-risk.git

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Recap: Project Overview

The Goal: Predict if a loan applicant will default. (Binary Classification)

Why It Matters:

- For Lenders: Reduce financial risk & losses.
- For Applicants: Increase fair access to credit.

Data: Kaggle's "Home Credit Default Risk" Competition.

- **Content**: Application data + Historical loan records.
- **Challenges:**
    - **Large Data**: Over 300,000 applicants and 134 features.
    - **Missing Data**: 50% features have missing values.

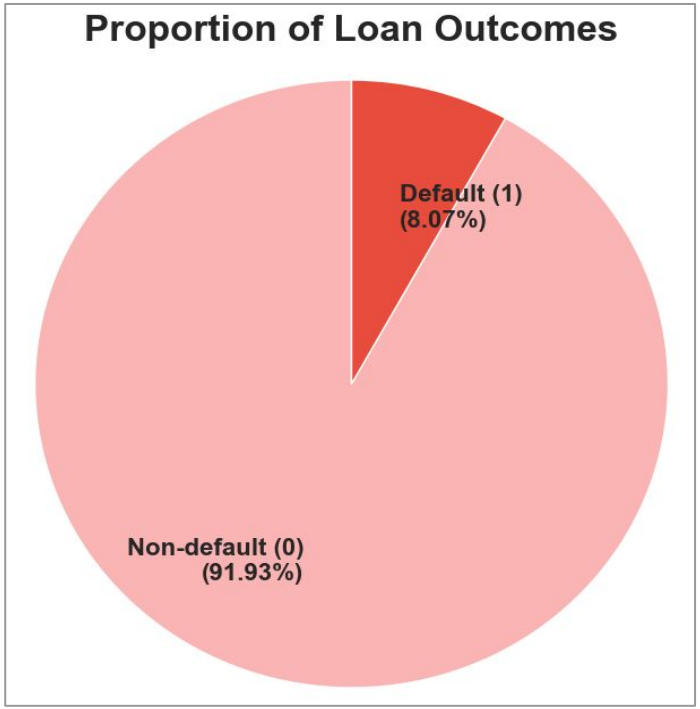
Method I: Splitting Strategy

Splitting Strategy:

- **Method**: Stratified Train-Validation-Test Split (60% / 20% / 20%).

- **Data Size**:

  Train (184,506) | Val (61,502) | Test (61,503)



Feature Selection:

Dropped non-predictive columns (e.g., Application ID, Gender) -> **134 features**

Method II: Preprocessing Pipelines

	Pipeline A: For Linear Models (LR, SVM)	Pipeline B: For Tree Models (XGB, RF, DT)
Ordinal Features	**Imputation:** Missing values treated as a new category. **Encoding:** Ordinal Encoding	Identical Strategy
Categorical Features	**Imputation:** Missing values treated as a new category. **Encoding:** One-Hot Encoding	Identical Strategy
Continuous Features	**Imputation:** Iterative Imputer using **Random Forest**. **Scaling:** Log-transformation (for skewed financial data) + StandardScaler.	No Preprocessing Required

Method III: Models & Hyperparameter Tuning

Input Space: *High-dimensional & sparse.*

- Original: 134 features -> **Encoded: 255 features**.

Handling Imbalance:

- Applied class_weight = 'balanced' (or scale_pos_weight) across all models.

Optimization Strategy:

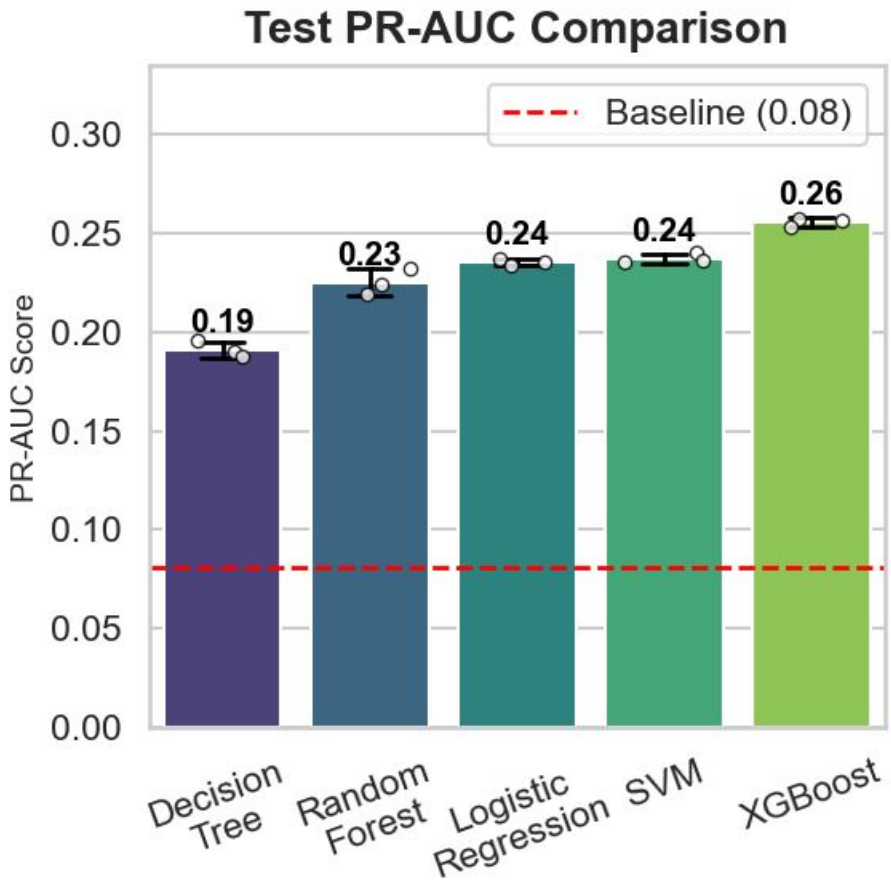
- Tuning via Hold-out Validation (Train-Val Split).
- Metric: **PR-AUC**.

Robustness Check:

- Repeated with 3 Random Seeds (42, 12, 25).

Model	Tuned Key Hyperparameters
Logistic Regression	C (Regularization), Penalty (L1/L2).
Linear SVM	C (Margin hardness).
Decision Tree	Max Depth, Min Samples Leaf, Max Features.
Random Forest	N Estimators, Max Features.
XGBoost	Learning Rate, Subsample, Depth; *Early_stopping = 50*.

Results I: Model Performance Comparison

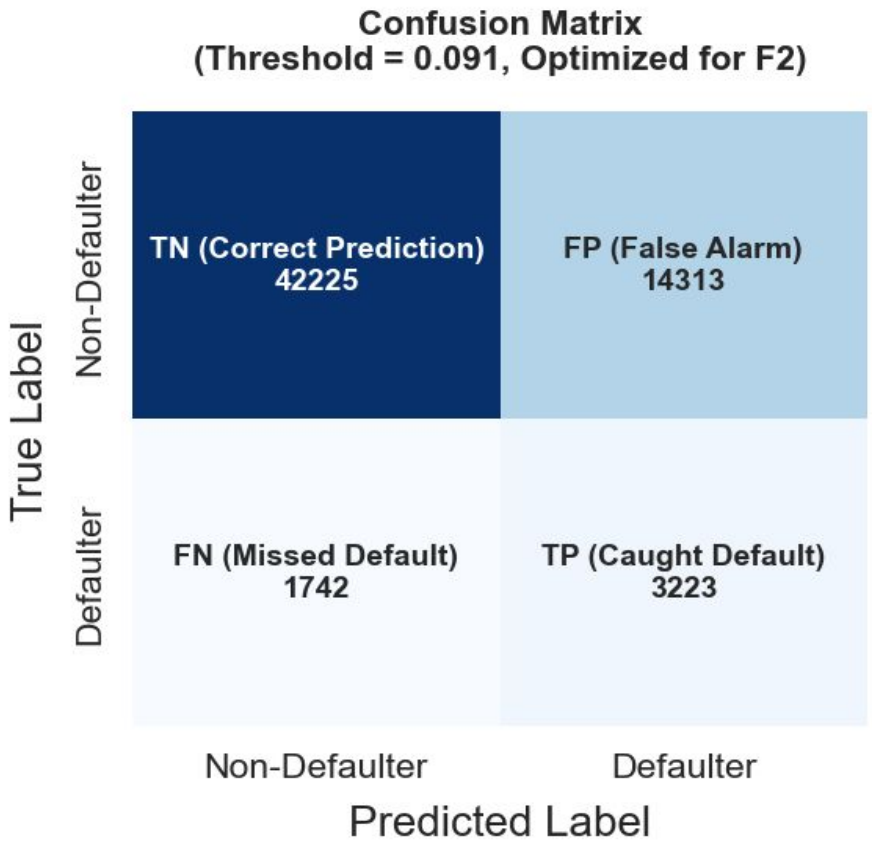


1. **The Baseline Model**
   - Predict all applicants as defaulters.
   - Score: 0.081 (rate of true positive class).

2. **The Winner: XGBoost**
   - The best performance (PR-AUC: **0.256**).
   - **~3x better** than the baseline (0.081).

3. **Statistical Significance:**
   - Standard Deviation = 0.002
   - **76 std deviations** above the baseline.

Results II: Business Impact & Decision Threshold



$$F_2 = (1 + 2^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(2^2 \cdot \text{Precision}) + \text{Recall}}$$

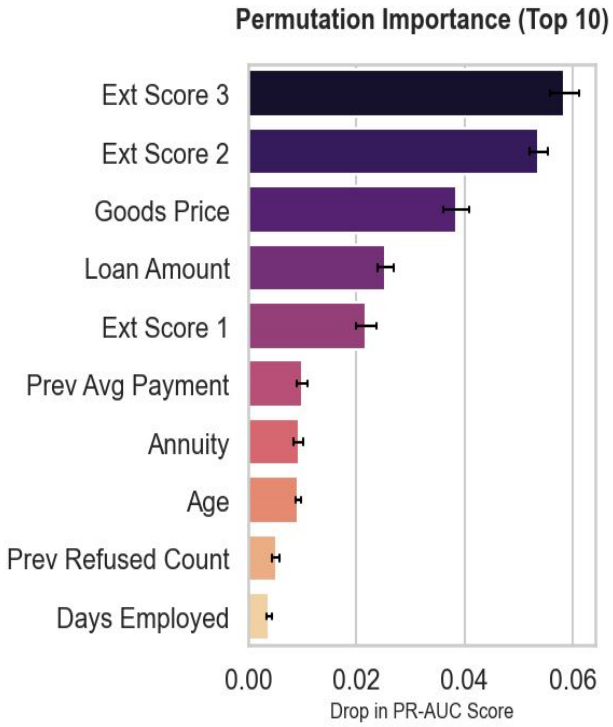
- **Strategy:** Optimized for F2-Score.

  $\beta = 2$ -> weights Recall higher.

- **Threshold:** 0.09 (Strict risk control).

- **Result:** Captured 65% of defaulters (Recall).

Interpretability I: Global Feature Importance



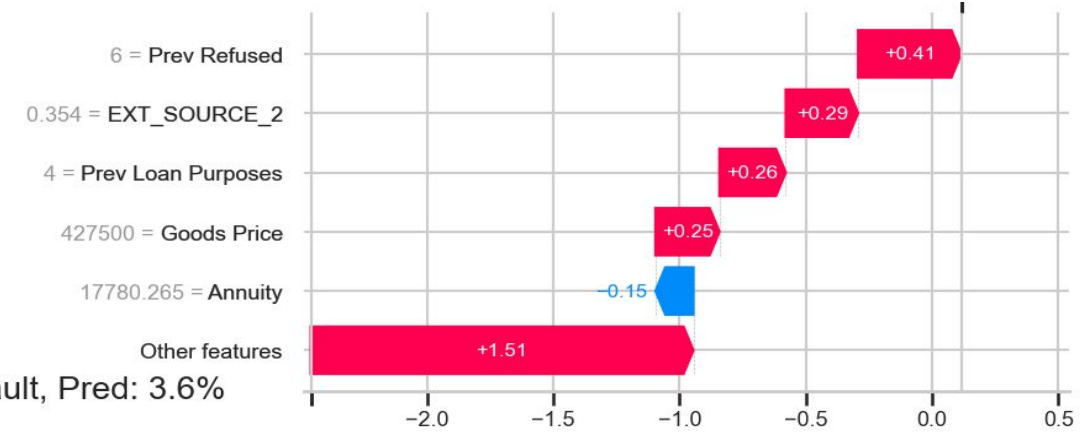
Driven Factors: ***External Credit Scores; Income Stability and Loan Magnitude.***

Interpretability II: Local Error Analysis

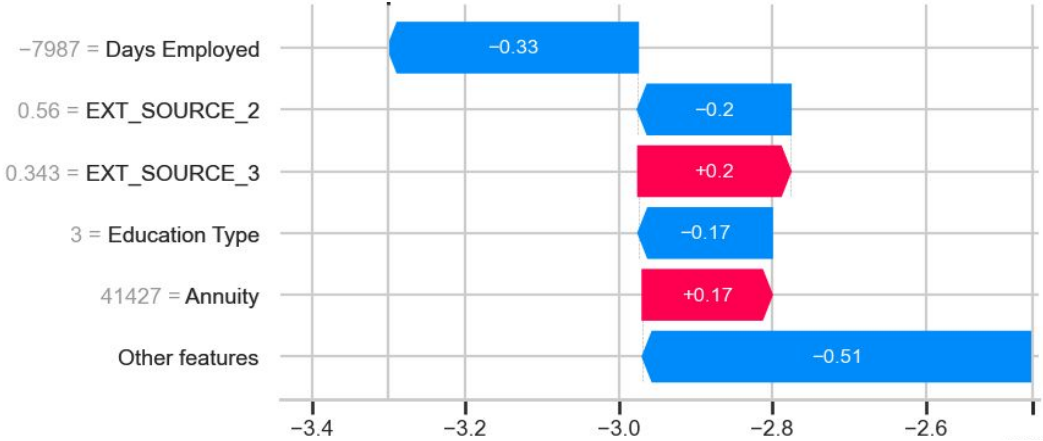
FP Case (False Alarm): ID 61931, True: Paid, Pred: 52.9%

The "False Alarm" Case

- **Key Drivers (Red Bars):**
  - `Prev Refused` **(+0.41)**
  - `EXT_SOURCE_2` **(+0.29)**



FN Case (Missed Fraud): ID 233233, True: Default, Pred: 3.6%



The "Missed Fraud" Case

- **Key Drivers (Blue Bars):**
  - `Days Employed` **(-0.33)**
  - `EXT_SOURCE_2` **(-0.20)**

Outlook: Limitations & Future Work

1. Advanced Sampling:

- Apply SMOTE to synthesize minority samples to address class imbalance.

2. Feature Engineering:

- Engineer Interaction Features to extract deeper insights from historical data.

3. Stacking Ensemble:

- Combine multiple models to leverage more signals for higher predictive power.

Thank You for Listening!

Any Questions?